Eg: Swiggy

SDLC (Software Development Lifecycle)

SDLC is a Step by Step procedure to develop and to deploy a Software

Stages of SDLC

1. Requirement Collection
2. Analysis/Feasibility Study
3. Design
4. Development
5. Testing
6. Deployment
7. Maintenance

1. Requirement Collection
   Collecting the requirement from the Users/endusers

2. Analysis/Feasibility Study
   Group of people are going to decide to take up this project or not

3. Design
   Creating a blueprint of the project based on the requirement

4. Development
   Based on the Design/Blueprint the developers will write the code

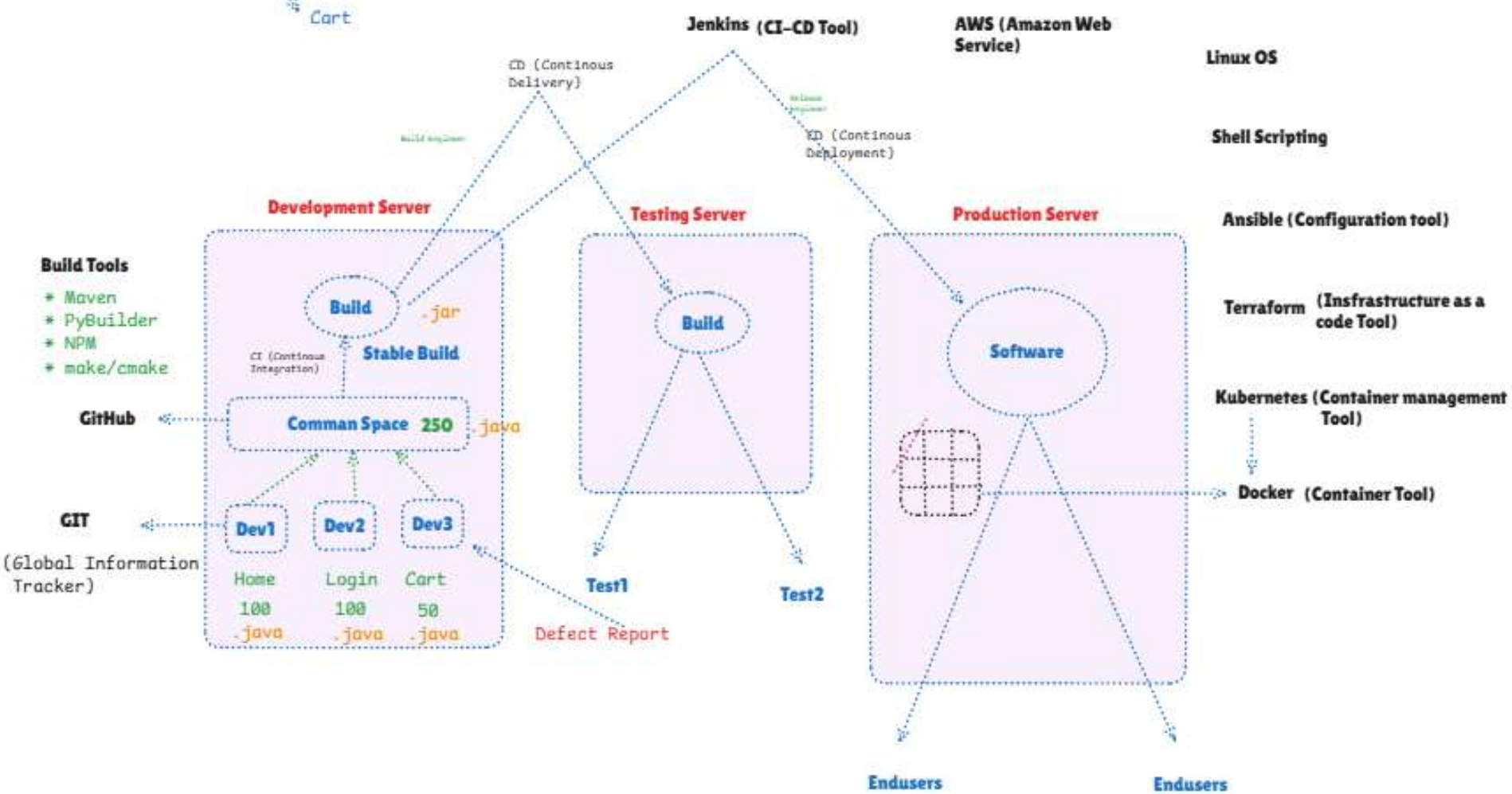
5. Testing
   After developing the Software it has to be tested in order to make sure the software is error free

6. Deployment
   This is where we are going to hand over the software to the customers

7. Maintenance
   Once after the deploying the software fixing the minor defects is called maintenance

DevOps

DevOps is a process/approach to develop and to deploy a software in a faster



Drawbacks of Traditional Approach

1. There is no collaboration between the developers
2. Chances of making the error is high
3. There is no collaboration between the developent team and the operation team
4. There is a delay in the Deployment of the Software

Stable Build:
Stable build is a Software with a lest defect

GIT (Global Information Tracker)

It is used to track the versions the source code i,e

you can see "who" , "what", "when"

* Github

It is a place where you are going to host your source code.

* Build Tools

It is used to convert source code into an executable/Build file
--> There are many build tools
* Maven --> Java
* PyBuilder --> Python
* NPM (Node Program Manager) --> JS/Node js
* make/cmake --> c/c++

* Jenkins (CI/CD tool) .jar (java archieve)

It is heart of DevOps, It is used to automate the complete IT Operations
--> CI (Continous Integration)
Converting the source code into an executable file (Build File) is called as Continuous Integration

--> CD (Continous Delivery)
Installing the build file from the development server to the testing server is called Continous Delivery

--> CD (Continous Deployment)
Deploying the stable build from the development server to the production server is called Continous Delivery

* Docker (Container Tool)

It is mainly used to mange the application in the productions server and Docker will provide the containers

will provide the containers

* Kubernetes (Container Management Tool)

  It is used to maintain and manage the Docker

* Terraform (Infrastructure as a code Tool)

  It is used to create, manage and automate the infrastructure of the servers

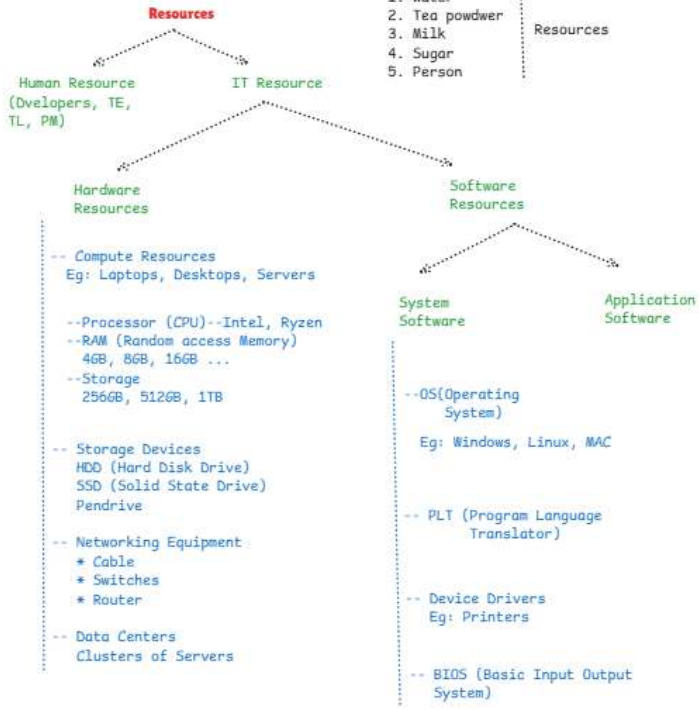
* Ansible

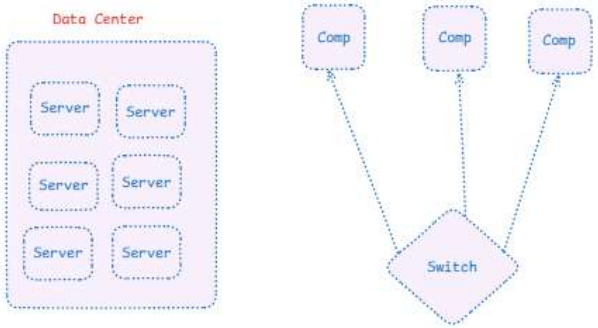
  It is used to configure/customize multiple Servers

Steps to prepare a Tea:

1. Water
2. Tea powdwer
3. Milk
4. Sugar
5. Person

Resources





System Software
It is a software used to manage the hardware

PLT (Program Language Translator)
It is going to convert one programming language to another programming Language

Device Drivers
It is going to help your OS to communicate with other hardware devices

BIOS (Basic Input Output System)
*It is a chip mounted on your motherboard.
* It is the first software which get started when you start/reboot your system

Application Software:
It is a software used to perform Specific Task
Eg: Netflix--> Watching Movies
Google chrome --> used for web browsing
MS word --> writing Documents

Switches
It is hardware device used to connect multiple devices in the same network

Router
It is used to connect multiple networks

Server
It is a big computer. It is a place where your applications are running

Traditional Approach Drawback

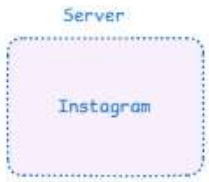
To move canvas, hold mouse wheel or sp

1. Initial investment is High

   Laptops--> 1 lakh --> 100*1= 1 crore
   Infrastructure
   Staff
   Network Euipments
   Storage Devices
   Servers---> 5 crores
   Data Centers

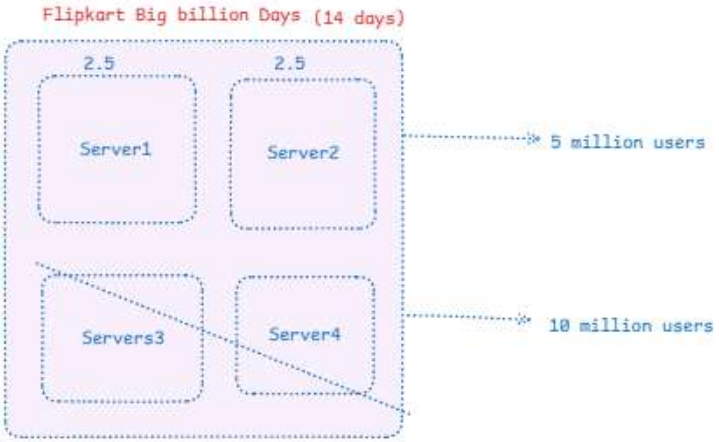
2. Maintaining the Servers is Diffcult

   24/7 the server has to be kept onn--> generate heat
   Power Supply
   AC
   Staff

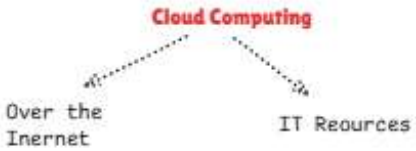


3. No Scalability or Less Scalability

   Increasing or decreasing the server is difiicult



4. Accessibility is Less



Cloud computing is a technology which allow the users to access the computing resources over the Internet
Eg: Servers, Networking Equipments, Databases, storage devices etc

Features of Cloud Computing

1. On-demand Self Service

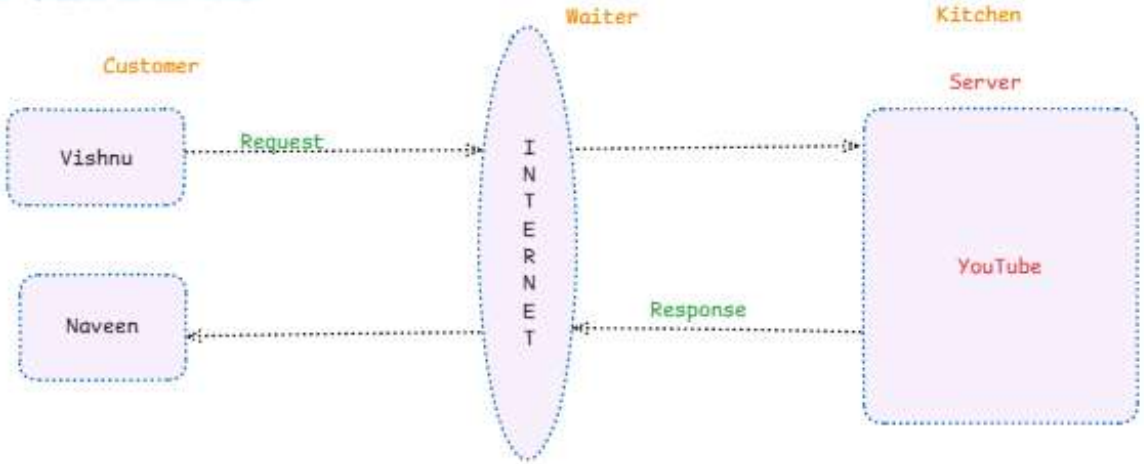
Purchase 100 Servers

You can get all the resources instantly
(Servers, Storage devices, Networking Equipments)

2. Pay as you go model (PayG)

Pay only for the resources what your going to use
Eg: Electricity bill

3. Client Server Model



Types of Cloud Computing

- Cloud Deployment Model
- Cloud Service Model

* Cloud Deploymnet Model

Cloud Deployment Model will define the cloud based on ownership, usage and purpose

Cloud Deployment Model

- Public Cloud
- Private Cloud
- Hybrid Cloud

Public Transport Eg: Government Bus	Private Transport Eg: CAR	Combination of Both Eg: Taxi
-- All the users can access the bus	-- Owners can access it	-- Entry Restricted
-- No customization	-- Customization is possible	-- Moderate Customization
-- Pay for the distance	-- Pay for the distance	-- Pay for the distance
-- Bus Company/Organization is going to take care of the maintenace	-- Owner of the car is responsible for maintenance	-- Owner of the car is responsible for the maintenance
-- Less security	-- High Security	-- Limited Security

-- Public Cloud

* Open for all the Users
* Customization is not possible
* It follows pay as you go model
* Organization (Third party cloud providers) is going to take care of the maintenance
* Less Secured compared to private Cloud

-- Private Cloud

* Authorized users has the permission
* Customization is possible
* It follows pay as you go model
* Authorized users is going to take care of the maintenance
* It is highly Secured

-- Hybrid Cloud

* Entry Restricted
* Moderate customization is possible
* It follows pay as you go model
* Owner (Third party cloud providers) is going to take care of the maintenance
* Moderate/Limited Security

Cloud Service Model

-- **IAAS (Infrastructure as a Service)**

-- **PAAS (Platform as Service)**

--**SAAS (Software as a Service)**

To move canvas, hold mouse wheel or spacebar while drag

*** Infrastructure as a Service** (Eg: Rented flat)

-- Servers	-- Water Supply
-- Storage	-- Electricity
-- Networking Equipments	-- Rooms

*** Platform as a Service** (Eg: Semi Furnished Flat)

-- Servers	-- Water Supply
-- Storage	-- Electricity
-- Networking Equipments	-- Rooms
-- OS (Operating System)	-- Comordes
-- Runtime Environment	-- Fans

*** Software as a Service** (Eg: Fully Furnished Flat)

-- Servers	-- Water Supply
-- Storage	-- Electricity
-- Networking Equipments	-- Rooms
-- OS (Operating System)	-- Comordes
-- Runtime Environment	-- Fans
-- Application	-- Furnitures
-- PL (Programming Language)	

Cloud Service Providers

1. AWS (Amazon Web Service)
2. Microsoft Azure
3. GCP (Google Cloud Platform)
4. Oracle Cloud
5. IBM Cloud
6. Alibaba
7. VMware
8. Digital Ocean
9. Dropbox
10. Rackspace

AWS Services Across various Domains

1. Cloud Financial Management
   Eg: Billing and Cost Management

2. Compute
   Eg: EC2 (Elastic Compute Cloud),Lambda

3. Storage
   Eg: S3 (Simple Storage Service), EBS (Elastic Block Store)

4. Database
   Eg: Amazon RDS, Dynamo DB

5. Networking Service
   Eg: VPC (Virtual Private Cloud), ELB (Elastic Load Balancer), Route53

6. App Integration
   Eg: SNS (Simple Notification Service)

7. Security, Identity and Compliance
   Eg: IAM (Identity and access management)

8. Management
   Eg: Cloud Watch

AWS (Amazon Web Service)

-- It is a subsidiary of Amazon
-- It was launched in the year 2002
-- SQS (Simple Queue Service), It was launched in the year 2004
   It is used to maintain the communication within the organization
-- In 2006 they officialy launched two different Services
   1) EC2 (Elastic Compute Cloud)
   2) S3 (Simple Storage Service)
-- They are providing 200+ different services accross the the globe/world

AWS Global Infrastructure

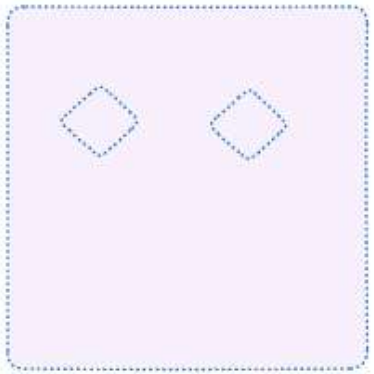
--> There are 4 different components of AWS Global Infrastructure
1) Regions
2) Availability Zone
3) Edge Locations
4) Regional Edge Caches

--> Regions
Regions are "Geographical Location" specified by AWS

--> Availability Zone
It is the location where your data centers are present

Min 3
Max 6

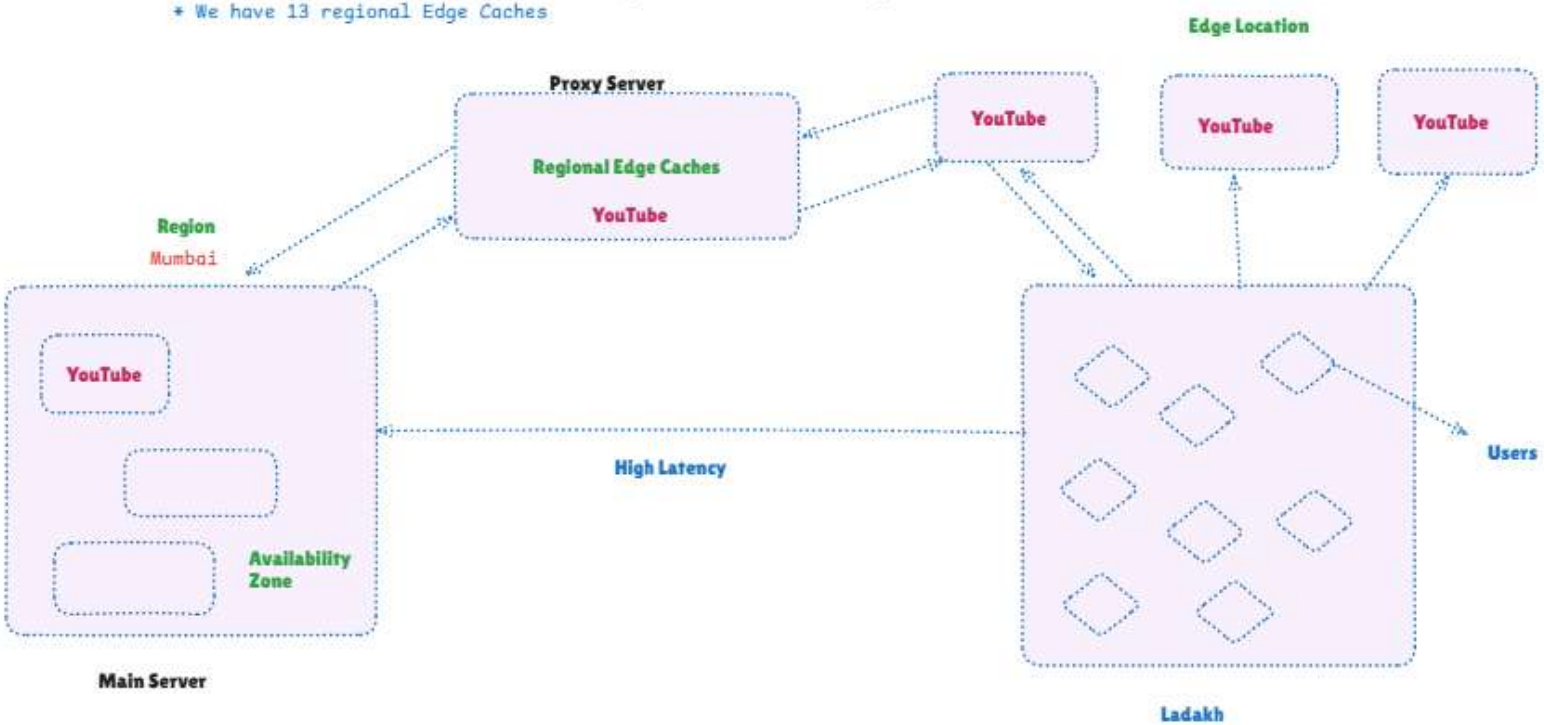
If we have minimum 3 availability zone and maximum 6 "AZ" then it is called as region



--> Edge Locations

These are the datacenters present outside the region
* It is also called as "Cloud Front POP's" (Point of Presence)
* Totally there are 700+ Edge Locations
* These are the datacenters which are present near to the users

--> Regional Edge Caches
These are the big data centers which is present outside the region
* We have 13 regional Edge Caches



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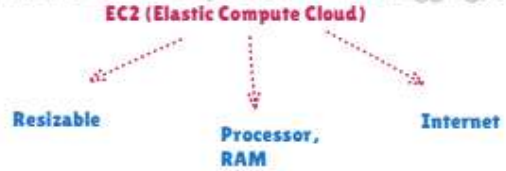


EC2 (Elastic Compute Cloud)

* It comes under Compute Category
* It is one of the web service provided by AWS
* It is going to provide you the virtual servers over the internet

Components of EC2

1. Instance
   It is nothing but your virtual Servers

2. EBS (Elastic Block Store)
   It is going to provide storage for the instance over the internet

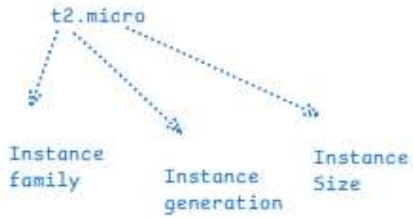
3. ELB (Elastic Load Balancer)
   It is used to distribute the load/users to multiple servers

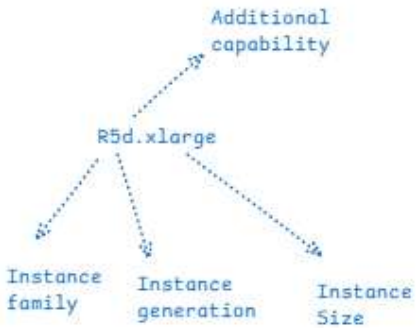
4. ASG (Auto Scaling Group)
   It is used to increase or decrease the number of resources based on usage

5. AMI (Amazon Machine Image)
   It is a blueprint of your Operating System for creating the Instance

Instance Types

1. General Purpose
2. Compute Optimized
3. Memory Optimized
4. Accelerated Computing
5. Storage Optimized
6. HPC (High Performance Computing)





* General Purpose
  It has a balance of compute, memory and networking
  family: t and M
  Eg: It is used to run small scale applications

* Compute Optimized
  Which requires high computational power that is which requires high CPU
  family: C
  Eg: High performance web servers

* Memory Optimized
  Application that requires more memory/RAM
  family: R and X
  Eg: Big data analytics, High end gaming

* Storage Optimized
  Application that requires more Storage
  family: I and D
  Eg: Store large amount of data

* Accelerated Computing
  It is a type of special hardware like GPU (Graphics processing unit) which is used to make certain task faster
  family: D,G,P and T
  Eg: High End Gaming

* High performance computing
  When you require high computing resources to solve complex problems that requires high processing power that is memory, Storage and networking
  family: H
  Eg: Trading

Key Pair

It is used to securely connect to your Intsance
There are two different types of Key-pair
1. Public key pair
2. Private key pair

There are two different extensions
1) .pem (Privacy enhanced mail)
2) .ppk (Putty private key)

Steps to create EC2 Instance

1. Go to AWS management console and search for EC2 and click on it
2. In Ec2 console, Select the Instance that is present at the left side
3. Click on Launch Instance, and name your instance
4. Select the OS (Amazon Linux)
5. Select the instance t2.micro
6. Create the Key pair
7. Select the Storage (8GB)
8. Click on Launch Instance

Steps to connect your EC2 instance

* Select the instance that you want to connect and click on connect

Steps to connect using SSH Client

1. Select the instance that you want to connect
2. Click on Connect
3. Select the SSH client (Secure Shell) option
4. Copy the examample commands
5. Go to downloads and command prompt from that location where your private key pair file is present
6. Paste the example command
7. You will be able to establish a connection

Steps to create and connect to RDP (Remote Desktop Protocol)

1. Click on Launch Instance
2. Select "Windows OS" and the default storage for windows is 30GB
3. Launch Instance
4. Select the created instance, Click on connect and Select RDP Client
5. Download Remote Desktop file and click on get password
6. Upload the private key and encrypt the password
7. Copy the password , run the remote desktop file and paste the password
8. Click on Connect

Basic Linux Commands

1. sudo su - (It is used to switch to the "root" user)

2. mkdir directory_name (It is used to create a directory)

3. ls (It is used to list all the files and directories)

4. touch file_name (It is used to create a file)

5. vi file_name (It is used to create a file and add content inside the file)

   Note:
   :wq --> write and quit
   :q! --> quit without making any changes

6. cat file_name (It is used to read the content inside a file)

7. nano file_name (It is used to create a file by adding content inside a file

8. clear (clear the screen)

9. cd (It is used to change the directory)

10. cd .. (It is used to move one step backward)

11. cd ../.. (It is used to move two step backward)

12. rm file_name (It is used to delete the file)

13. rmdir directory_name (It is used to delete an empty directory)

14. rm -r directory_name (It is used to remove the directory as well as the files inside the directory)

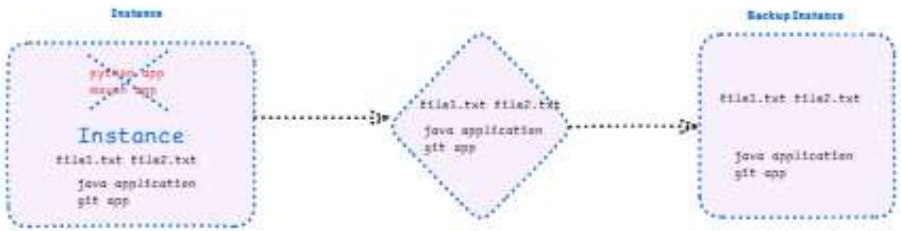
Commands to install the Software

1. sudo yum update (It is used to update the package)
2. sudo yum install java -y (It is used to install java)
3. sudo yum install python -y (It is used to install python)
4. sudo yum install git -y (It is used to install git)
5. sudo yum install maven -y (It is used to install maven)
6. java --version (To check the java version)
7. python --version (To check the python version)
8. git --version (To check the git version)
9. mvn --version (To check maven version)
10. sudo yum remove "software_name" -y (It is used to remove the software)

AMI (Amazon Machine Image)

AMI is a template/blueprint which contains software configurations like Operating System and application

There are two different types of AMI
1. Default AMI (That is provided by third party providers)
2. Custom AMI



Using AMI

1. You are going to backup instance
2. You can create a template to launch multiple Instances at the same time

There are two ways to get backup for your EC2 instance
1. AMI (Amazon Machine Image)
2. Snapshot

Steps to create AMI

1. Select the instance that you want to create a backup
2. Click on actions and select image and template
3. Select create image
4. Give name and description
5. Click on create

Steps to create instance using AMI

1. In AWS EC2 dashboard search for AMI in the left side
2. Select the AMI from which you want to launch a new instance and click on launch instance
3. Give name for your instance
4. Click on Launch Instance

Snapshots

It is used to get the backup of your volume/storage

Steps to create AMI using Snapshot

1. In EC2 dashboard Select snapshot
2. Select the snapshot from which you want to create an AMI
3. Click on Actions, Then select create image from snapshot
   --> Image will be created
4. Using that image launch an Instance

EBS (Elastic Block Store)

* It is one of the storage service provided by AWS
* It will provide block level storage (volumes)
* Volumes are availability zone specific
* Instances are region Specific
* Volumes are also called as Block level storage

There are two different types of volumes

1. Instance Store Volume
2. EBS volume

1. Instance Store Volume

It is a temporary block level storage and it will be created when you launch an instance' and it will be deleted when you terminate the instance

2. EBS Volume

These are the volumes created by users and it has to be deleted by the users itself

Steps to create a volume

1. Go to volume
2. Click on create volume
3. Change the size of the volume
4. Change the availability zone based on which you have launched the Instance because volumes are availability zone specific
5. Click on create volume

Steps to attach volume

1. Select the volume which you want to attach
2. Click on action--> attach volume
3. Select the instance that you want to attach

Steps to create volume and attach it to windows Operating System

1. Create a Instance using Windows OS
2. Create a Volume and attach it to the instance
3. Connect the instance using RDP (Remote Desktop Protocol)
4. In search bar, type "Server Manager" and wait for few minutes you will get an option named "File and storage Service" at the top left corner
5. Click on File and Storage Service
6. You will get an option named "Disks"
7. Right click on the volume which you have created--> Initialize
8. Right Click--> New Volume
9. Click on Next--> Next--> Create
10. Go to file explorer--> Click on "This PC" and check the disk partition.

Steps to Create Volume and Mount in Linux Operating System
* By performing this you will be able get the backup of the volume using snapshot

1. Create an instance with Linux OS
2. Create a volume and attach it to the instance
3. Connect to the instance

Commands:

1. lsblk (List information about the block devices)

2. sudo fdisk -l (list information about all availables storage devices in a long listing format)
   copy the path Eg: /dev/xvdd

3. sudo file -s /dev/... [path of volume] (to verify the file system)

4. sudo mkfs -t xfs /dev/.. [path of volume] (It is used to create a file system)

5. sudo mkdir directory_name (It is used to create a directory)

6. sudo mount /dev/...[path of volume] directory_name

7. lsblk

It will be mounted

S3 (Simple Storage Service)

* It comes under storage category
* It is one of the storage service provided by AWS
* S3 provides a scalable storage
* In s3 files are stored in the Objects
* S3 is global service
* Objects are stored inside the bucket

--> Buckets
Buckets are the containers for the data stored in S3

--> Objects
Objects are the fundemental entities stored in s3

EBS (Elastic Block Store)	S3 (Simple Storage Service)
1. It provides storage for EC2	-- It is an independent service
2. It is block level storage	-- It is object storage
3. It is Availability zone specific	-- It is a global service
4. The maximum volume capacity 16 TB	-- It is going to provide you unlimited storage

Steps to create a bucket

1. Go to S3--> Click on create Bucket
2. Give the bucket name according to the naming rules (The bucket name must be unique)
3. Go with the default options
4. Click on Create bucket

Steps to upload a file/Object

1. Get inside the bucket
2. Upload the objects
3. Click on Upload button

Note: Before deleting the bucket make sure you are going to empty the bucket

Steps to host a static html page using S3 bucket

1. Create a bucket with ACL's
2. Uncheck block public access
3. Acknowledge it once after making the changes
4. Create Bucket
5. Once after uploading the object
   --> Copy the object URL and paste it in the browser--> You will get an error called access denied
6. Go to permissions--> Under ACL's click on edit and give permission for OBJECT and OBJECT ACL
7. Click on save changes
8. Refresh the page to get the output

Steps to perform static web hosting using S3

1. Create a bucket
2. Go to tooplate.com in the browser and download any of the template
3. Extract the files which you have downloaded
4. Go inside the bucket and upload the files and folders into the bucket
5. Click on Upload button
6. Click on properties--> Scroll down and click on edit the static website hosting
   --> change it to enable--> save changes
7. Copy the endpoint and paste it in the browser (You will get the page as access denied)
8. Go to permission--> Bucket policy--> Click on policy generator
9. Select s3 policy
10. In prinicipal give *
11. Under action select the "Get Object" option
12. Give the ARN (Amazon Resource Name) --> arn://...../*
13. Click on Add statement--> Generate policy--> Copy the json format and paste it in the bucket policy
14. Click on Save changes and refresh the page

Bucket Versioning

Bucket versioning means having multiple variants/versions of the object in the same bucket is called as bucket versioning

Bucket Replication

S3 is a replication feature in Amazon s3 that automatically copies object from one bucket (source) to another bucket(destination)

Steps to perform bucket versioning

1. Create 2 bucket by enabling the bucket versioning
2. Go to management in the source bucket--> Click on create replication rule
3. Give the replication rule name--> Browse S3 and select the destination path --> Create a new role (to access from one bucket to another bucket we need to create a role)--> Click on Save
4. Check whether the bucket replication has been enabled--> If it is enabled--> Upload the object in the source bucket and that will be reflected in the destination bucket also.

IAM (Identity and access management)

* It comes under the category of Security, Identity and Compliance management
* It is one of the web service provided by AWS
* IAM is a global service
* It is used to create users, usergroups, policies and roles

It is a service provided by AWS. It allows you to manage users, usergroups, policies and roles to define who can access what within AWS account.

Users

It is an identity with long term credentials that is used to interact with AWS in an Account

Steps to Create Users

* Go to IAM Service
* Click on Users
* Click on create user--> give the user name--> Provide access to AWS management console
* Select the custom password
* Uncheck the box inorder to avoid creating new password whenever you login
* Click on Create user
* Once after creating user--> Copy the console signin URL and paste it in the new window and give the username and password

Policies
Ploicies defines the permissions
There are two different types policies

1.AWS managed (default)
2.Customer managed

Steps to create policies

1. Select the policies
2. Select the required policies--> Click on actions
3. Attach policies to the user

Steps to create customer managed policy

1. Select the policies--> Create policy
2. Select a service which you want to access (eg: s3, EC2)
3. Allow all actions and click on next--> Scroll down under resources select all
4. Give the policy name and click on create policy
5. If you want to see the policies which you have created
   --> In policies--> Under Filter by type cahange it to customer managed and select the policies which you have created.

User Groups

User group is a collection of IAM users. It is used to specify permission for a group/collection of users.

Steps to create user group

1. Click on user groups
2. Click on create group
3. Add user to the group (Note: You should create a user)
4. Attach the policies to the group
5. Click on create user group

Roles

IAM Role is an identity with short term credentials. You can create Roles that has specific permissions that are valid shorter duration.

Note: When you want to access from one service to another service then create a role.
Eg: (Ec2 to S3)

Steps to create IAM Role

1. Navigate to IAM Service
2. Click on roles under access management
3. Click on create role
4. Trusted entity type as default (AWS Service)
5. Usecase--> Select EC2
6. Select the policies --> EC2fullaccess
7. Give the name for the role and create the role

Verify S3 from EC2

--> aws s3 ls (To list the buckets)

--> aws s3 mb s3://bucket_name (It is used to create/make bucket)

--> aws s3 cp object_name s3://bucket_name (It is used to upload an object inside the bucket)

--> aws s3 ls s3://bucket_name (To list the objects inside the bucket)

--> aws s3 rm s3://bucket_name/object_name (To delete files/objects from the bucket)

--> aws s3 rm s3://bucket_name/--recursive (To delete all the objects at one shot)

--> aws s3 rb s3://bucket_name (To delete the bucket)

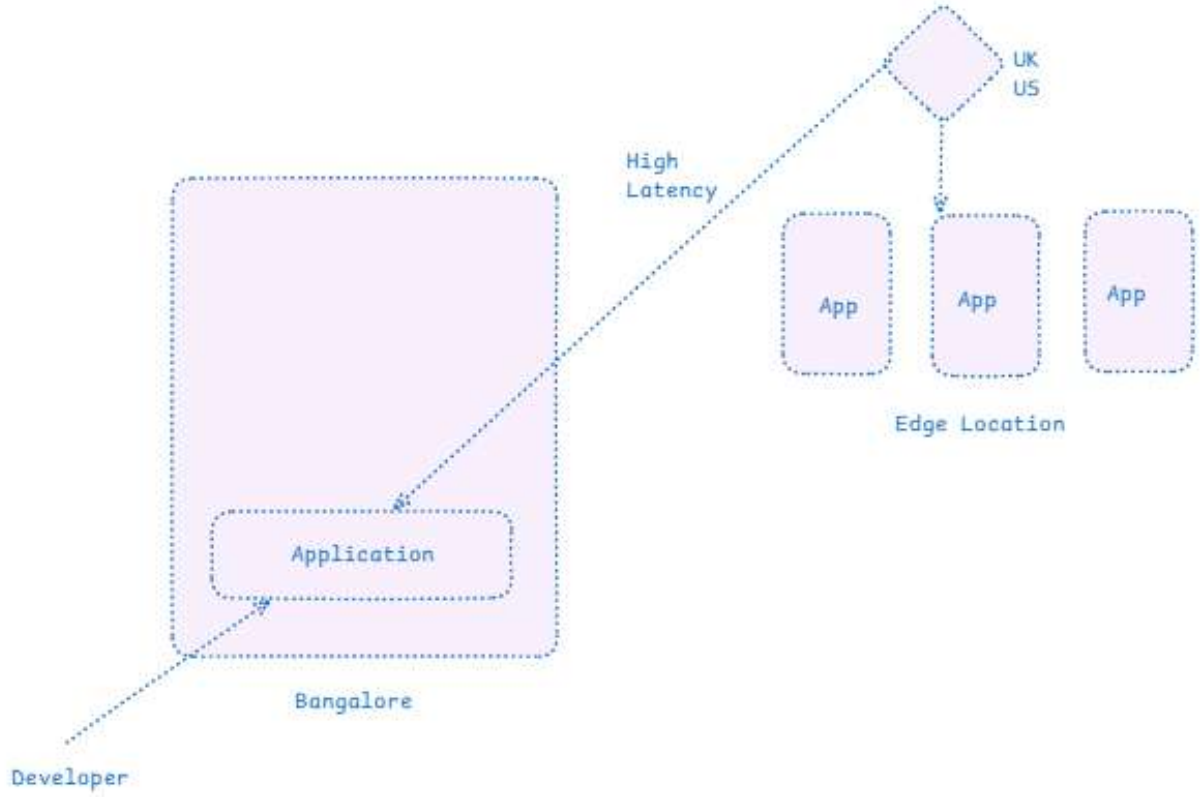
TASK-1

1. Create 5 users
2. Create a usergroup of 4 users
3. Create a policy (Customer managed)--> Create ec2 and s3 fullaccess
4. Attach 2 policies to the group
5. Launch instance and modify IAM role
6. Upload 4 objects in s3 using EC2

CloudFront/CDN (Content Delivery Network)

AWS CloudFront is a Content Delivery Network (CDN) service provided. It enables fast and secure distribution of content such as

* Web pages
* Videos
* Images



Steps to perform CloudFront

1. Perform static web hosting using S3
2. Go to CloudFront Service
3. Click on Create Distribution--> Choose the bucket under origin domain--> Under Origin access select the Legacy access identity--> Create new OAI (Origin access identity)--> Click on I want to update the bucket policy--> WAF (Web application firewall) enable this--> Price class --> use all edge locations--> Under default root object give it as "index.html"
4. Click on Create distribution
5. Copy the DNS and paste the DNS in the browser

Deploy an application in the Web Server

Operating System

* Linux
* Windows
* Ubuntu
* mac OS
* Kali Linux

Port numbers:

* http: 80
* https: 443
* ssh: 22
* mysql: 3306
* Jenkins: 8080
* Tomcat: 8080
* Postgresql: 5432

Package managers:

* amazon linux - yum
* ubuntu - apt

Servers/engines

* nginx
* httpd
* apache

Steps to deploy an application using nginx server (Linux OS)

(Default path of nginx server in Linux OS: /usr/share/nginx/html)

* Launch an instance using Linux OS
* Connect to the EC2 terminal
* Go to tooplate.com --> Select a template --> Right click on the download button --> Copy the link address
* In the EC2 terminal type the command that is

  --> wget -O file_name.zip url(Paste the copied url)

* Unzip the file

  --> unzip file_name

* Install the nginx server

  --> sudo yum install nginx -y

  --> nginx -v (Check, the server is installed or not)

  --> systemctl status nginx (To check the status of nginx)

  --> systemctl start nginx (To start the nginx server)

* Copy the public IP and paste it in the browser--> you will be not able to access the site--> Check the port number--> make changes in the security i.e give http --> Youu will get "Welcome to nginx page"

* Go to the EC2 terminal--> To deploy the inside the nginx server use this command

  sudo cp -r directory_name/* /usr/share/nginx/html

* Once after deploying the application in the nginx server--> Refresh the page you will be able to get the deployed web application page

Steps using Ubuntu OS

(Default path for ubuntu OS: /var/www/html)

* Launch an instance using Ubuntu OS
* Connect to the EC2 terminal
* Update the packages using the command---> sudo apt update
* Go to tooplate.com --> Select a template --> Right click on the download button --> Copy the link address
* In the EC2 terminal type the command that is

```
--> wget -O file_name.zip url(Paste the copied url)
```

* Unzip the file using the command --> you will get a error so install unzip first using the command

```
--> sudo apt install unzip

--> unzip file_name (To unzip the file)
```

* Install the nginx server

```
--> sudo yum install nginx -y

--> nginx -v (Check, the server is installed or not)

--> systemctl status nginx (To check the status of nginx)
```

* Copy the public IP and paste it in the browser--> you will be not able to access the site--> Check the port number--> make changes in the security i.e give http --> Youu will get "Welcome to nginx page"

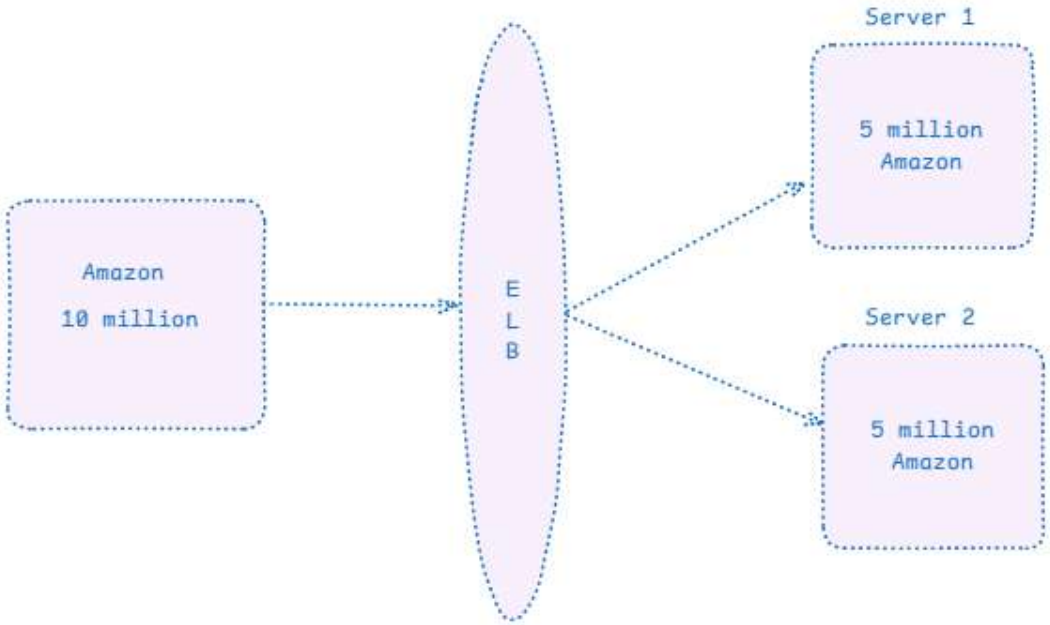
* Go to the EC2 terminal-->
  To deploy the inside the nginx server use this command

```
sudo cp -r directory_name/*  /var/www/html
```

* Once after deploying the application in the nginx server--> Refresh the page you will be able to get the deployed web application page

ELB (Elastic Load Balancer)

--> It comes under networking category
--> Elastic load balancer is one of the web service provided by AWS cloud and it will distribute the load(incomming traffic) to the given target (Instances)



Target Group

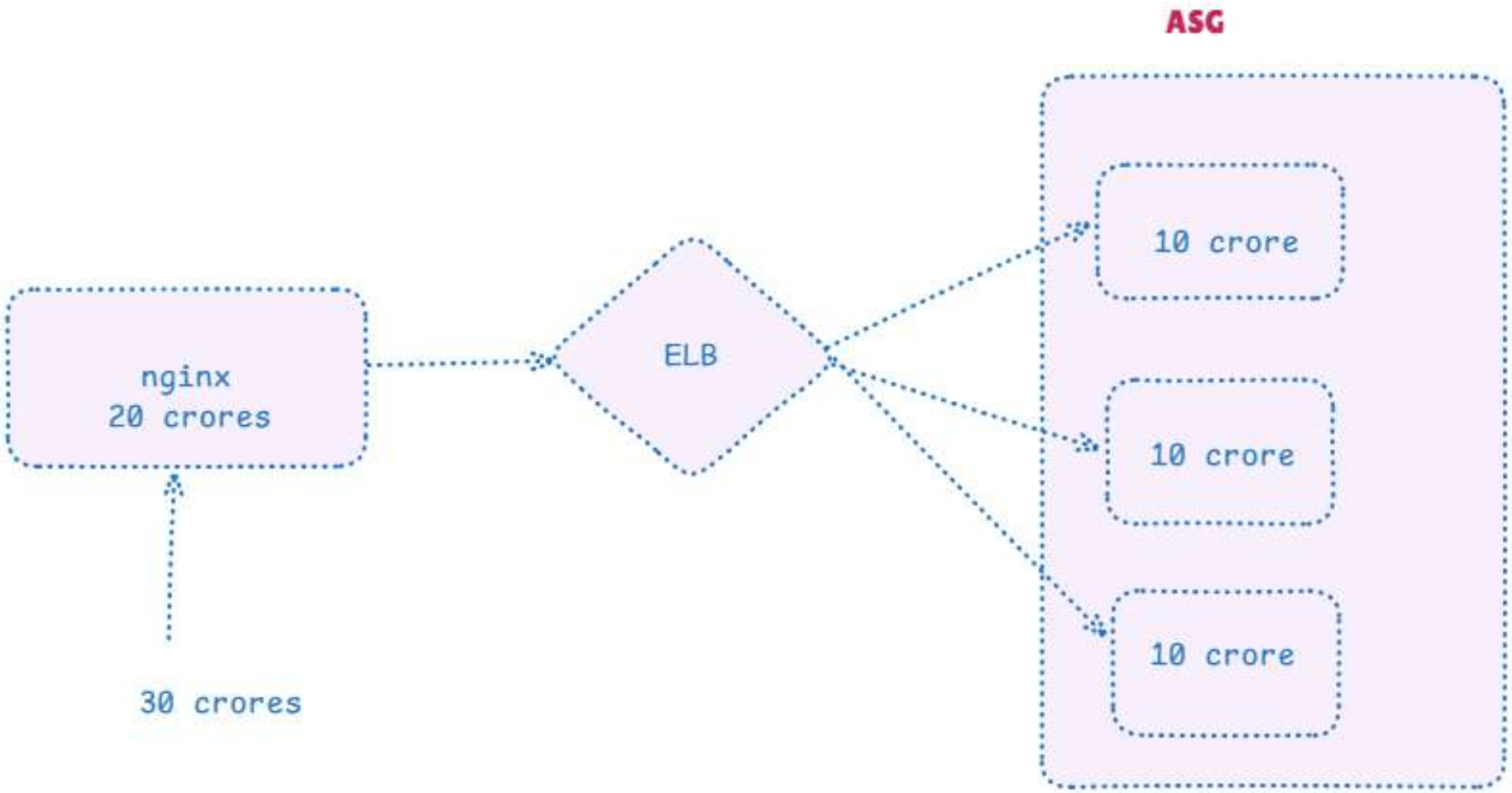
It is a way to manage and route the traffic to multiple EC2 instances or other AWS services when using a ELB

Steps to perform ELB

* Launch two instances with nginx installed--> Host a webpage in both the server
* Click on traget group--> Give the target group name--> Click on next--> Select the instance that you want to divide the load--> Click on include as pending below--> Click on create target group
* Click on load balancer--> Create load balancer--> In application load balancer click on create
* Give load balancer name--> Select minimum two availability zone
  In default action select target group
* Click on craete load balancer
* Paste the DNS (Domain name system) in the browser

ASG (Auto Scaling Group)

Autoscaling group is a collection of ec2 instances that enables automatic scaling



Steps to perform ASG

* Launch an Instance using Linux OS
* Install nginx and start the nginx server
  Commands:
  --> sudo yum install nginx -y
  --> systemctl start nginxfac
  --> systemctl enable nginx
* Go to tooplate--> copy the webpage to nginx default path that is /usr/share/nginx/html
  Note: Deploy a static webpage in the nginx server
* Create a backup of the instance that is AMI
* Once the AMI is active--> Stop the instance
* Click on Launch Template--> Give the template name--> Give the version
  --> Check the autoscaling guidence option--> Select the AMI which you
  have created
  --> Select the keypair, instance type(t2.micro), select the Security group
* Click on Autoscaling group --> create Autoscaling group--> Select the template
  --> Click on next--> Select minimum 2 availability zone--> next
  --> Attach new load balancer--> Internet facing --> Create a target group--> Next
* Desired Capacity (Eg: 3) Maximum capacity (5)--> Click on target tracking scaling
  policy--> CPU utilization (set it as 70) --> Next-->Create AutoScaling Group
* Go to load Balancer --> Copy the DNS and paste it in the browser-->
  You will be able to get the web page

Steps to delete the resources

Delete the AutoScaling group--> Delete the Load balancer--> Delete the target groups
--> Delete the AMI--> Snapshots --> Template --> Instances

To move canvas, hold mouse wheel or spacebar while dragging, or use the hand tool

AWS CLI (Command Line Interface)

It is a tool to manage AWS services using Command Lines

There are 3 ways to access your AWS console

1. Web-based AWS console (GUI Login)
2. AWS Mobile APPlication
3. AWS CLI

Steps to perform AWS CLI

1. Install AWS CLI Installer from the browser
2. Go to the command propmpt --> configure the cmd using the commands (aws configure)
3. Provide
   * AWS access key
   * Secret Key
   * Default Region
   * Output format go with default one just press enter

To get the access key and Secret Key--> Create a user--> Go inside the user--> Click on create accesskey --> Copy the access key and the secret key

Commands:

1. type nul > file_name (It is used to create a file)
2. dir (To list all the directories and files)
3. dir /a:-d (To list only the files)
4. del file_name (to delete the files)

EC2 Instance

--> To List the EC2 instances--> aws ec2 describe-instances

--> Create an EC2 instance--> aws ec2 run-instances --image-id your-ami-id --instance-type t2.micro --key-name your-key-pair-name

--> Stop EC2 instances--> aws ec2 stop-instances --instance-ids your-instance-id

--> Start EC2 instances--> aws ec2 strat-instances --instance-ids your-instance-id

--> Terminate EC2 instances--> aws ec2 terminate-instances --instance-ids your-instance-id

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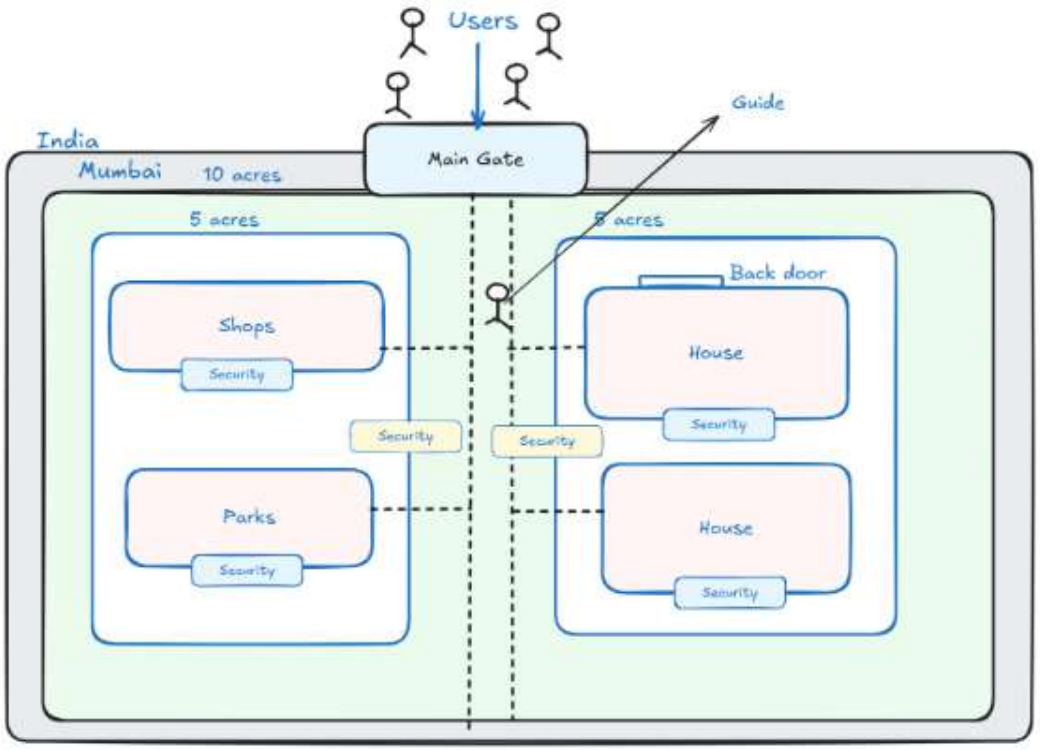
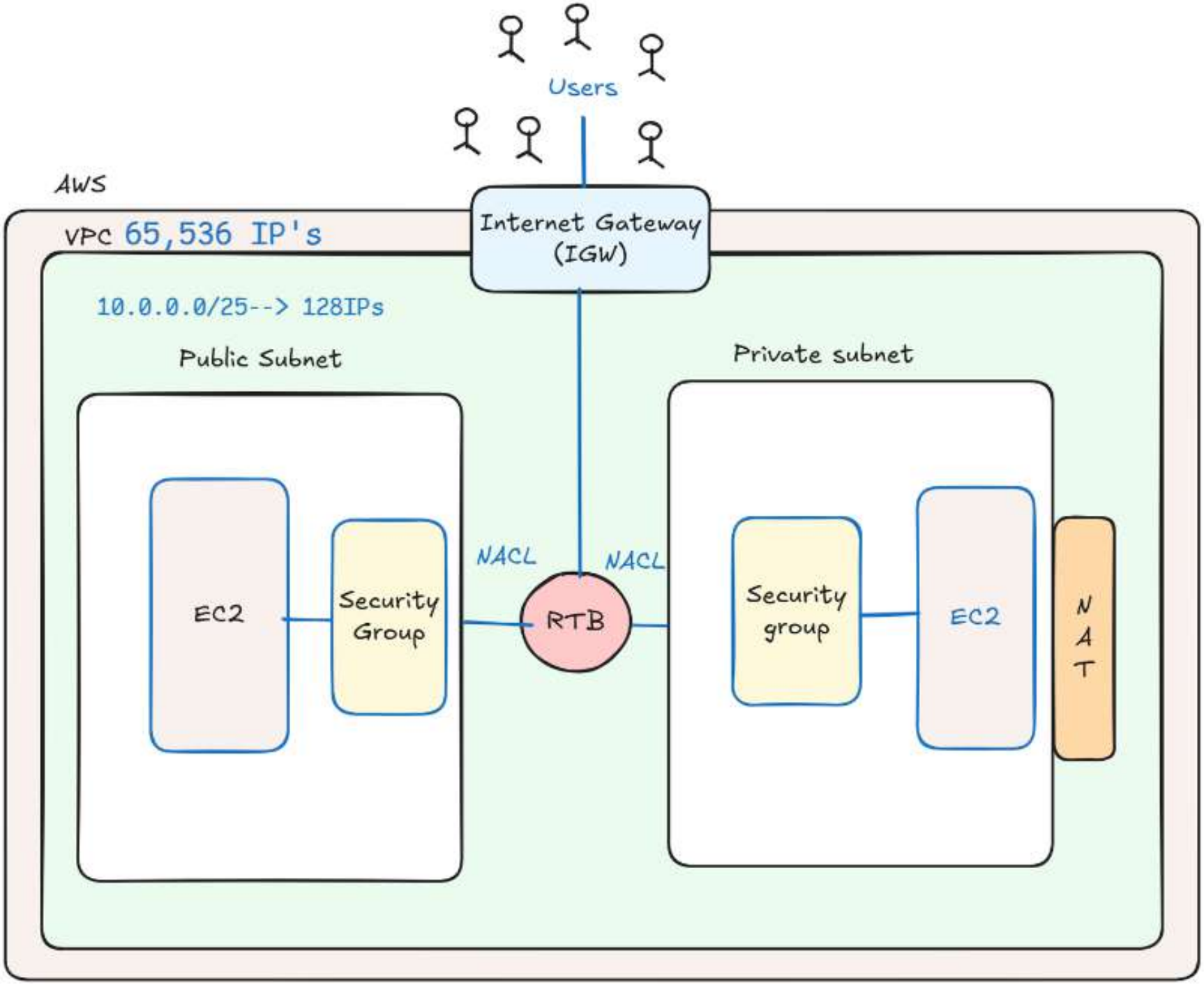
VPC (Virtual Private Cloud)

* It is one of the webservice provided by AWS
* VPC allows you to create isolated resources in the private cloud

There are two different types of VPC)
1. Default VPC
2. Custom VPC (Managed by the customers)




Users
AWS
VPC 65,536 IP's
Internet Gateway
(IGW)
10.0.0.0/25--> 128IPs
Public Subnet
Private subnet
EC2
Security
Group
NACL
RTB
NACL
Security
group
EC2
N
A
T

* VPC (Virtual Private Cloud)
  It is one of the service provided by AWS. It allows you to create isolated resources in the private cloud

* Subnets: It is small portion of your VPC
  --> Public Subnet (Host resources that need internet access like websites)
  --> Private Subnet (Contains sensitive information like database) that should not be directly accessed over the internet

* Internet Gateway
  It is a component that enables the communication between the internet and the VPC

* Security Groups
  It will control inbound and outbound traffic (users) at the instance level

* NACL (Network Access Control List)
  It will control inbout and outbound traffic at the subnet level

* Route Table (RTB)
  It will direct the users/traffic to the subnets

* NAT Gateway (Network address translator)
  It will allow only the outbound rule

To move canvas, hold mouse wheel or spacebar while dragging, or use the hand tool

Address

Physical Address

Logical Address

Physical Address	Logical Address
--> Physical address are generated by the hardware	-- Generated by the Software
--> It is static	-- It is Dynamic (It keeps on changing)
--> Not discoverable over the internet	-- It is discoverable over the Internet
--> It is also called as MAC address (Media access control)	-- It is called as IP address
--> Command to check the physical address getmac	-- Command to check the logical address ifconfig

IP Addresses (Internet Protocol Address)

IPV4 (Internet Protocol Version 4)

IPV6 (Internet Protocol Version 6)

--> IPV4 (Internet Protocol Version 4)

10.0.0.0/20 --> 4096

Protocol:
It is set of rules

8bit 8bit 8bit 8bit = 32bit

4096/255=17

--- . --- . --- . ---

10.0.0.255

1 st place is occupied

0-255 0-255

Octet
Octet is an
8 bit number

10.0.1.255
10

0.0.0.1
0.0.0.2
.
.
.
0.0.0.255
.
.
.
0.0.1.0

0-255

255.255.255.255

256^4 = 4,29,49,67... = 4.3 billion users

Eg: 129.11.75.239

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
1	0	0	0	0	0	0	1
0	0	0	0	1	0	1	1
0	1	0	1	0	0	1	1
1	1	1	0	1	1	1	1

Binary format

10000001.00001011.01010011.11101111 --> 129.11.75.239

Eg: 123.15.82.255

* Classes of IPV4 Addresses

--> Class A: Range of class A is 0.0.0.0 - 127.255.255.255
Eg: It is used by internet service providers

--> Class B: Range of Class B is 128.0.0.0 - 191.255.255.255
Eg: It is used in medium organisation

--> Class C: Range of Class C is 129.0.0.0 - 233.255.255.255
Eg: It is used for small corporation

--> Class D: Range of Class D is 224.0.0.0 - 239.255.255.255
Eg: It is used for multicasting

--> Class E: Range of Class E is 240.0.0.0 - 255.255.255.255
Eg: It is kept for experimental purpose
Walkie-talkie

IPV6(Internet protocol version 6)

16bits

___ : ___ : ___ : ___ : ___ : ___ : ___ : ___ = 128bits

0-65,535

0-9
a-f
Hexadecimal number

0,1,2,3,4,5,6,7,8,9
a,b,c,d,e,f

65,536^8= 340 trillion users

3.4*10^8

IPV4 (Internet Protocol Version 4)

--> It has 4 Octets

--> Each Octet has 8 bits

--> The total number of Octet is 32 bits

--> It is seperated by dot (.)

--> The range is between 0-255

--> It can be used by 4.3 billion users

IPV6 (Internet Protocol Version 6)

--> It has 8 Octets

--> Each Octet has 16 bits

--> The total number of Octet is 128bits

--> It is seperated by colon (:)

--> The range is between 0-65,536

--> It is used by 340 trillion users

Steps to delete the resources

1. Delete the NAT Gateway
2. Subnets
3. Route Table
4. Detach the Internet Gateway from your VPC--> Internet Gateway
5. Delete VPC
6. Delete "Elastic IP" --> (Release the Elastic IP)

Steps to create VPC Only

--VPC

1. Give the VPC name
2. Give the IPV4 CIDR block --> 16--> 65,536 IP's will be allocated to your VPC
3. Tenancy --> Default
4. Click on Create VPC

--Subnets

1. Click on subnets
2. Select the VPC ID for creating the subnet
3. Give the subnet name
4. Select the availability zones of the subnet
5. Click on Create Subnet

--RouteTable

1. Click on route table
2. Give the route table name
3. Select the VPC
4. Click on Create Route Table

To move canvas, hold mouse wheel or spacebar while dragging, or use the hai

--Connect RouteTable to subnets

1. Select the subnet which you want to change
2. Click on Actions
3. Click on Edit Route Table Assosciation
4. Select the Routetable ID
5. Click on Save changes

--Internet Gateway

1. Click on Create Internet Gateway
2. Give the Name for the Internet Gateway
3. Click on Save Changes

--Attaching Internet Gateway to the VPC

1. Select the Internet Gateway which you want to attach
2. Click on Actions
3. Attach to VPC

--Attach Internet Gateway to the route table

1. Go to routes
2. Select the route which you want to connect to the internet gateway
3. Click on Actions
4. Click on edit routes
5. Click on add routes
6. Give the value as 0.0.0.0/0 in destination
7. In target select the internet gateway
8. You need to select the internet gateway which you have created
9. Click on Save changes

--NAT Gateway

1. Click on NAT Gateway
2. Give the name
3. Select the subnet (Public subnet)
4. Allocate the Elastic IP
5. Click on create NAT gateway

--Attach NAT gateway to route Table

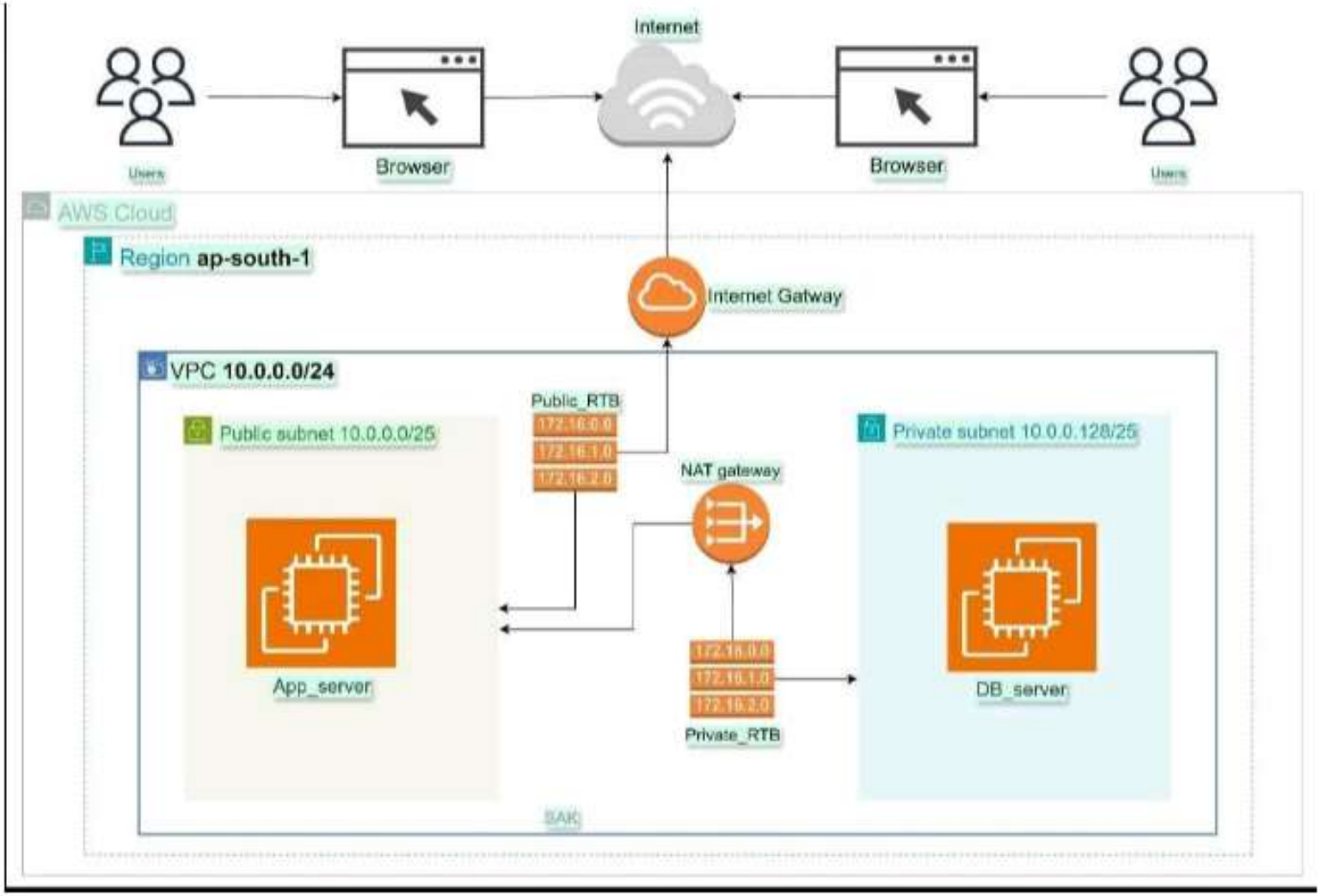
1. Go to routes
2. Select the route which you want to connect to the internet gateway
3. Click on Actions
4. Click on edit routes
5. Click on add routes
6. Give the value as 0.0.0.0/0 in destination
7. In target section select NAT Gateway
8. You need to select the NAT Gateway which you have created
9. Click on Save Changes

Note: Lauch a instance based on the Architecture

To move canvas, hold mouse wheel or spacebar while dragging, or use the hand to

TASK-1

VPC Architecture



Public Subnet: 10.0.0.0/25
Private Subnet: 10.0.0.128/25
VPC: 10.0.0.0/24

Cloud Watch

It is a monitoring service provided by AWS that helps you to track and manage AWS resources and services

Components of Cloud Watch

1. Metrics
2. Dashboards
3. Alarms

1. Metrics

--> milk = litres, ml
--> rice = kg
--> eggs = dozens

It collects the numericals data points
(Eg: CPU usage, memory usage, disk usage)

A metrics is something that helps you to understand about the utilization or the details of your AWS services

2. Dashboard

Custom visualization of the metrics in a single view
--> Using this dashboard you can analyze the resources based on Metrics

There are two different types of Dashboard
1. Custom Dashboard
2. Automated Dashboard

3. Alarms

time --> 6:00am --> Alarm should ring

Metrics condition Action

Similarly

CPU Utilization ---> Less than or equal to 30%---> Instance should stop running

Metrics	Condition	Action

Steps to perform Cloud Watch

1. Launch a Instance
2. Serach for cloud watch service
3. Go to Dashboard
   --> Click on create dashboard
   --> Give Dashboard name
   --> Click on Create Dashboard --> Select the metrics --> Click on next
   --> Select EC2--> Per-Instance metrics--> Select the CPU Utilization of that particular instance--> Create Widget
4. Go to Alarm--> In Alarm--> Click on Create Alarm
   --> Select metrics--> EC2--> Per-instance metrics--> CPU Utilization--> Select the treshold value (Lower than 40%)--> Next
5. To get the notification --> Create new topic--> Give the topic name and your email address where you want to receive the notification
   Note: Confirm the subscription
6. Click on EC2 actions
   --> Click on Add EC2 action --> In alarm --> * Stop instance
   * Terminate instance
   * Reboot the instance
7. Select the action which you want to perform
8. Click on Next
9. If the CPU Utilization exceeds the set limit the instance will get stopped and you will receive the notification

Lambda

* It comes under the compute category
* Lambda is a serverless compute category service which is being offered by AWS where you dont need to worry about the server configuration
* You need to write the code

OR

AWS is a serverless compute service that lets you to run the code without managing the servers

It supports multiple programming languages
Eg: java, .net, python, nodejs, ruby

EC2 (Elastic Compute Cloud)	Lambda
--> You need to create and manage the servers	--> No need to create and manage the servers
--> For scaling you have to setup autoscaling	--> Automatically scales based on the request
--> Pay only for running Instance	--> Pay only for execution time (milisecond)
--> We have to choose the Operating System, storage and software configuration	--> All these things will be taken care by the AWS itself

Lambda Pricing

Based on the "request per month"

1. First 1 million request per month are free
2. After that, you have to pay around $0.20 per million request

Based on "GB per second"

1. 400k GB-sec of computing time is free as per free tier account

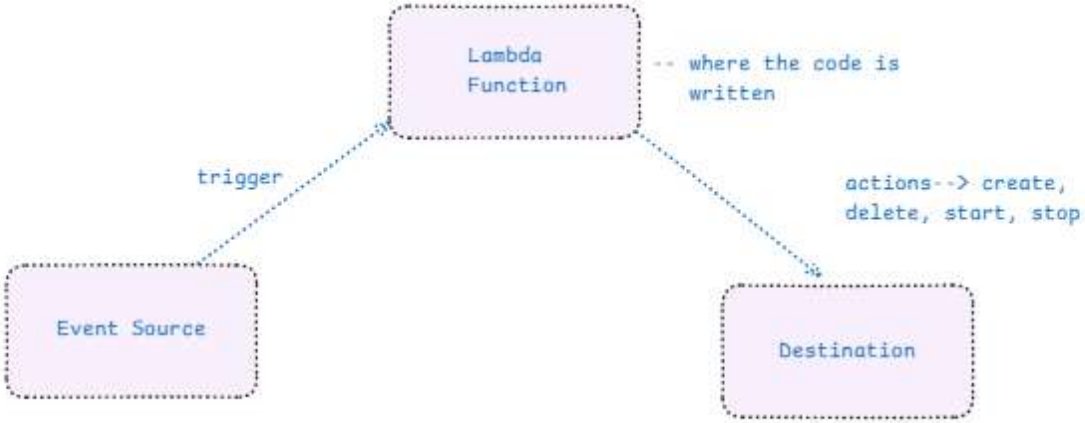
Lets say your code is taking--> 10sec
--> Code is comnsuming 100MB
10sec*100MB=1000

1GB=1024MB

1GB-sec

0.016$ per GB-sec

Structure of Lambda



Eg: SNS

(Ssimple Notification Service)

--> Function : The actual code that runs
--> Event Source: The trigger that starts the function
--> Destination: Where the result goes after execution

Types of Event Source

Push based event

Pull based event

--> Push based event

AWS automatically pushes the event to lambda function and the function executes immediately when the event occurs

--> Pull based event

Lambda pulls data from the event source
* The function executes only when the new data is available

Push based

Synchronous

Asynchronous

--> Synchronous
The event source waits for lambda to complete the execution
--> Synchronous event waits for lambda to execute the task before triggering another task

--> Aysnchronous
The event source doesnot wait for lambda to finish the execution
--> This event will trigger the lambda function immediately

--> Steps to Perform Lambda

1. Go to function
2. Select "Create a function"
3. Select author from scratch
4. Give the function name--> Select programming language (Python)
   --> Create function
5. Once your function is created go to code section
   --> Delete the default code and paste the script
6. Click on Deploy--> Test --> Create a new test event--> Event name--> Save
   --> Select the event--> Test

Note: you will get permission denied --> Create a role

7. Go to IAM Service--> Roles--> Usecase (Select Lambda)--> In permissions
   (Give EC2-fullaccess)
8. Give rolename--> Create role
9. Go to Lambda-->Configuration--> General Configuration--> Scroll down--?
   In existing role select the role which you have created by giving
   ec2fullaccess permission
10. Click on Save
11. Now deploy the code and Test

Script 1--> Start the instance
Script 2--> Stop the instance
Script 3--> It will stop the instance if it is running and it will start
            the instance if it is stopped
Script 4--> It is used to stop multiple Instances
Script 5--> It is used to start multiple Instances
Script 6--> It is used to terminate the stopped Instances

Canary Function

Canary function is a small test function that runs automatically at regular interval to check if a system or service is working fine

Steps to perform "Canary Function"

1. Create a function
2. Use a blueprint--> Search for canary function (schedule a periodic check)--> Select this
3. Go with the default options--> Create a new role
   --> Give the rule name--> Schedule expression --> "rate(1 minute)"

   Note: if you are passing more than 1 minute it should be in minutes

4. Click on Create function
5. Go to SNS Service--> Create a Topic--> Add subscription and confirm the subscription
6. Add destination--> Select the topic which you have created and click on save
7. You will receive the notification

SNS (Simple Notification Service)

* It comes under the app integration category
* It is a messaging service provided by AWS that enables the applicatons,services and users to send the notification

Topic:

It is a logical communication channel that allow the publishers to send messages to multiple subscribers at the same time

Steps to create SNS Topic

1. Search for SNS in AWS management console and select it
2. Click on create topic
3. Name the topic
4. Create a subscription
   --> Click on create subscription
   --> Select the protocol as email
   --> Give the e-mail ID and save it

Steps to add s3 as a trigger for Lambda

1. Create a function--> Author from scratch
2. Go to s3 service, create a bucket
3. In lambda service--> Click on "add trigger" --> Select the s3 which you have created
4. Select the bucket and select the event as "PUT"
5. Click on Save

Steps to attach SNS to Lambda Function

1. Go to Lambda function
2. Click on add destination
3. Select the SNS topic
4. Select the Topic Name
5. Click on Save

RDS (Relational Database Service)

* It comes database category

Database

A database is structured way to store and organize data so that it can be accessed easily

Types of Database

1.Relational Database
2.NoSQL Database

* Relational Database
  Store the data in the tables(like Excel) and it structure query language
  Eg: SQL, Postgresql, MySQL

* NoSQL Database
  Store data in a flexible way (like document or Key value pairs)
  Eg: Dynamo DB, Mango DB

To create a database

1. DBMS
2. Query Language
3. Host

* DBMS
  DBMS is a software that stores, organize and manage data in a structured way

* Host
  It is a device that connects to a network and communicate with other devices

QUERIES

1. SHOW DATABASES; ---> (It is used to list all the databases)

2. CREATE DATABASE DATABASE_NAME; ---> (It is used to create the database)

3. USE DATABASE_NAME; ---> (It will use the particular database which you want)

4. CREATE TABLE TABLE_NAME (
   ID INT AUTO_INCREMENT PRIMARY KEY,
   NAME VARCHAR(10),
   AGE INT); ---> (It is used to create a table)

5. INSERT INTO TABLE_NAME (NAME,AGE)
   VALUES
   ("xyx", 12),
   ("XYZ", 15); ---> (It is used to insert the values insude the table)

6. SELECT * FROM TABLE_NAME ---> (It is used to list all the details inside the table)

Dynamo DB

* It comes under compute category
* It will store the data in the form of "key" and "value" pairs
* It is unstructured data

Steps to perform Dynamo DB

1. search for Dynamo DB in the AWS management console
2. Click on create table
3. Give the table name
4. Give the partition key for eg: id
5. Click on create the table will be created
6. Go inside the table which you have created
7. Click on actions--> Click on create item to insert values into the table
8. If you want to add a new coloumn--> Click on add new attribute
9. It will create the attribute